


CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

Test Pattern

JEE(Advanced)
FULL SYLLABUS
08-05-2024
JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

Time : 3 Hours

PAPER-1 (OPTIONAL)

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY
GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

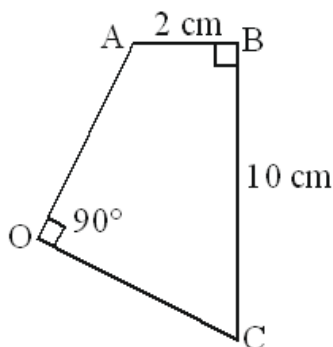
Negative Marks : -1 In all other cases

1. A taut string is made of two segments. To the left of point 'A', it has linear mass density of μ kg/m and to the right of A, its linear mass density is 4μ kg/m. A sinusoidal wave pulse is travelling towards right.



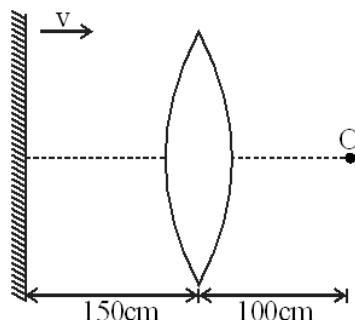
Which of the following is correct ?

- (A) Ratio of amplitude of reflected and transmitted wave is $\frac{1}{4}$
- (B) Ratio of amplitude of reflected and transmitted wave is 2
- (C) Ratio of transmitted power and reflected power is 8
- (D) Ratio of transmitted power and reflected power is 16
2. One mole of a monoatomic gas is heated in such a way that its molar specific heat is $2R$. During the heating, the volume of the gas is doubled. By what factor does the temperature of the gas change ?
- (A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$ (C) 2 (D) 4
3. The figure shows a uniform slab of mass 7 kg. Find the moment of inertia about the axis normal to the plane of drawing and passing through the point O. Given $OA = OC$, $BC = 10$ cm and $AB = 2$ cm.

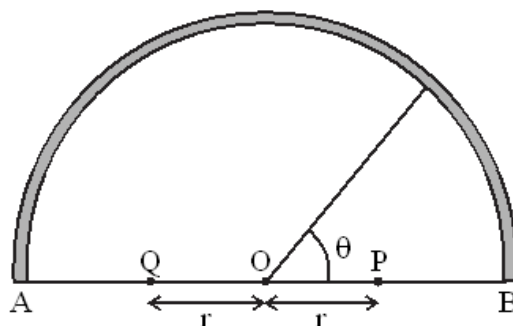


- (A) 84 kg cm^2 (B) 168 kg cm^2 (C) 188 kg cm^2 (D) 198 kg cm^2

4. An observer is standing at a point O, at a distance of 100 cm from a convex lens of focal length 50 cm. A plane mirror is placed behind the lens at a distance of 150 cm from the lens. The mirror now starts moving towards right with a velocity of 10 cm/s. What will be the magnitude of velocity of her own image as seen by the observer, at the moment when the mirror just starts moving ?



- (A) $\frac{200}{9}$ cm/s (B) $\frac{200}{3}$ cm/s (C) $\frac{20}{9}$ cm/s (D) $\frac{10}{9}$ cm/s
5. Charge distribution on a rod bent as a semicircular arc of radius R follows a function $\lambda = \lambda_0 \sin \theta$. Here λ is line charge density at a point on the rod, λ_0 is a positive constant and θ is angular position of the point as shown in the figure. V_0 , V_P and V_Q represents the electrostatic potential at points O (centre point), P and Q respectively as shown in figure. Which of the following is correct?



- (A) $(V_P = V_Q) < V_0$ (B) $(V_P = V_Q) > V_0$
 (C) $V_P = V_Q = V_0$ (D) $V_P > V_Q > V_0$
6. Nuclei of radioactive element A are being produced at rate t^2 at time t. The element A has decay constant equal to λ . If N is the number of active nuclei of element A at any time t then $\frac{dN}{dt}$ was found to be minimum at $t = t_0$. Find N at time $t = t_0$.

- (A) $\frac{2t_0 - 2\lambda t_0^2}{\lambda^2}$
 (B) $\frac{2t_0 - \lambda t_0^2}{\lambda^2}$
 (C) $\frac{t_0 - \lambda t_0^2}{\lambda^2}$
 (D) $\frac{2t_0 - \lambda t_0^2}{2\lambda^2}$

SECTION-I(ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

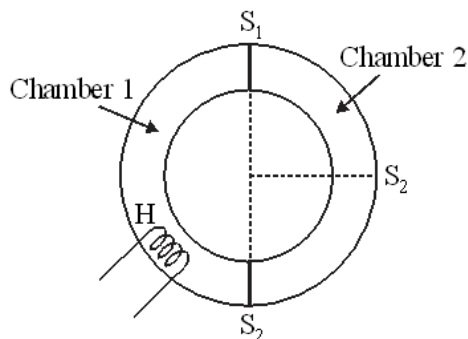
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

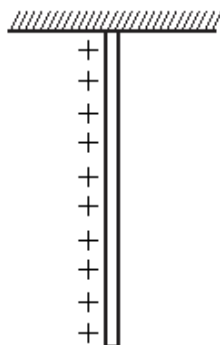
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. A ring shaped tube has uniform cross sectional area and its entire volume is $2V_0$. The tube is well insulated from the surrounding. Inside the tube there is an adiabatic fixed wall S_1 and another movable adiabatic partition S_2 . Initially, the movable partition is diametrically opposite to S_1 and the two halves of the tube have equal amount of an ideal gas ($\gamma = 1.5$) at same pressure P_0 . Now, a heater H is switched on which supplies heat slowly to one of the chambers. Heater is kept on till the partition S_2 moves through the quarter of the circle. At this position the heater is switched off and the partition S_2 remains in equilibrium. Neglect any friction as well as heat loss to the surrounding through the walls of the tube. Which of the following options are correct ?

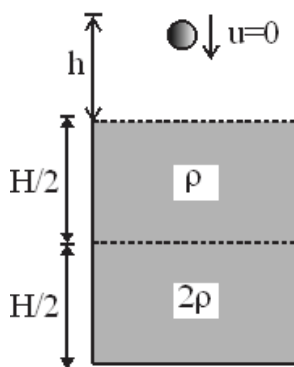


- (A) Final pressure in chamber '2' is $2P_0$
- (B) Final pressure in chamber '2' is $2\sqrt{2}P_0$
- (C) Work done on the gas in chamber '2' by the gas in chamber '1' is $(2\sqrt{2} - 1)P_0V_0$
- (D) Heat supplied by the heater to the gas is $4P_0V_0(2\sqrt{2} - 1)$

8. A uniformly charged non conducting rod is suspended vertically at its end. The rod can swing freely in the vertical plane without any friction. The linear charge density on the rod is λ C/m and it has a uniformly distributed mass of M . Length of the rod is L . A uniform horizontal electric field (E) is switched on in the region. Select the correct option/options :-

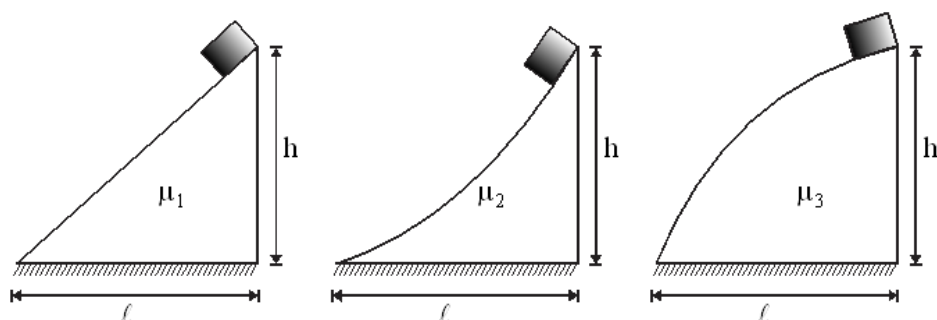


- (A) If $E = \frac{mg}{\lambda L}$, then rod just manages to make it horizontal
- (B) If $E = \frac{mg}{2\lambda L}$, then rod just manages to make it horizontal
- (C) If $E = \frac{Mg}{\lambda L}$, the rod has maximum kinetic energy at an angle 45° with vertical
- (D) If $E = \frac{Mg}{2\lambda L}$, the rod has maximum kinetic energy at an angle 45° with vertical
9. A small ball of mass m is released from rest from a height h above the liquid surface. The density of ball is $\frac{\rho}{3}$: (assume no splash)

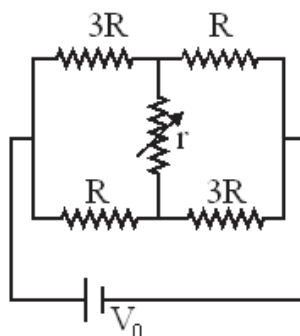


- (A) If $h \geq H$, then the ball will be able to enter the liquid of higher density
- (B) If $h \geq \frac{7H}{2}$, then the ball will be able to strike the bottom of the vessel
- (C) If $h = 2H$, then the ball will cross the interface of two liquids with kinetic energy = mgH
- (D) If $h = 0$ then ball will float on the surface.

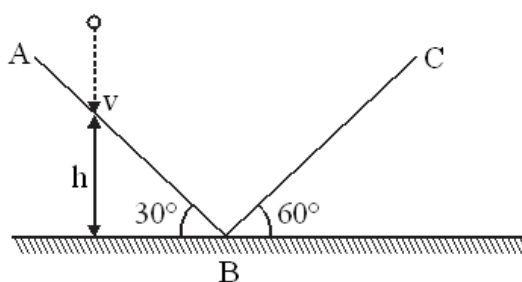
10. Consider three fixed surfaces shown in the figure. Three blocks each of mass m are released from rest from top of the three surfaces. All blocks reach ground with same speed. Length of path travelled by the blocks is same for second and third surface. If coefficient of friction of three surfaces are μ_1 , μ_2 and μ_3 respectively then



- (A) $\mu_1 = \mu_2$ (B) $\mu_1 > \mu_2$ (C) $\mu_2 = \mu_3$ (D) $\mu_2 < \mu_3$
11. In the circuit shown in figure R and V_0 are constant, while r can change, taking up all values. What can be the equivalent resistance of the circuit ?



- (A) $1.6 R$ (B) $1.4 R$ (C) $1.8 R$ (D) $2.2 R$
12. There are two inclined planes, as shown in the figure. A ball strikes the inclined plane AB vertically with velocity $v = \sqrt{24g}$ at height $h = (1 + 2\sqrt{3})$ m. The collision is inelastic. After this the ball moves such that it collides with the inclined plane BC perpendicular to it. Select the option(s).



- (A) The co-efficient of restitution between the inclined plane AB and ball is $\frac{1}{\sqrt{3}}$
- (B) The co-efficient of restitution between the inclined plane AB and the balls is $\frac{1}{3\sqrt{3}}$
- (C) The time interval between two collision is $\sqrt{\frac{8}{g}}$ s
- (D) The velocity of collision with the inclined plane BC is $\sqrt{6g} \left(1 + \frac{1}{\sqrt{3}} \right)$ m/s

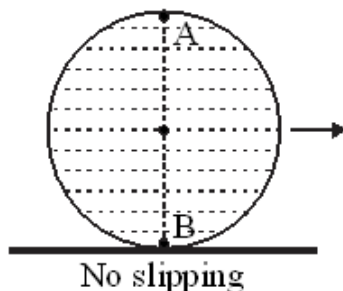
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

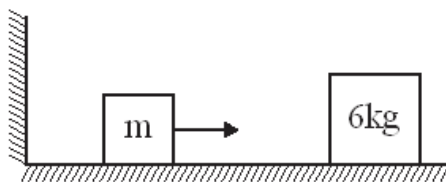
Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. A non-viscous incompressible liquid of mass m is filled fully inside a thin uniform spherical shell of mass m and radius R performing pure rolling on a rough horizontal surface. There are two point A and B inside the liquid on the vertical diameter separated by $2R$ as shown in the figure. Pressure difference between B and A is $(P_B - P_A)$. At the given instant velocity of centre of mass of given system is V_0 and kinetic energy of this system is K . If the value of $\frac{K}{P_B - P_A}$ is equal to $\frac{n\pi V_0^2 R^2}{9g}$. Find n .

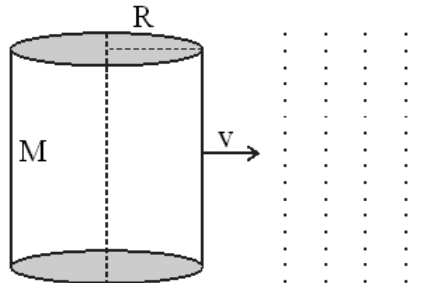


2. A block of mass ' m ' is given velocity towards another block of mass 6 kg kept at rest on a frictionless horizontal floor some distance apart from a wall as shown in the figure. If all collisions are elastic, find minimum value of mass ' m ' (in kg) so that blocks will collide only once.

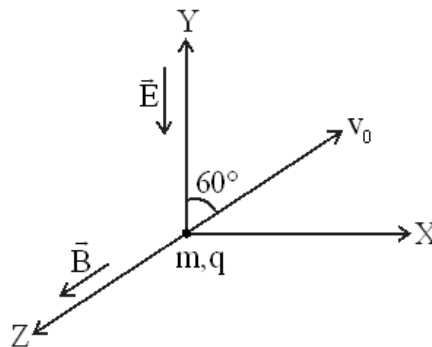


3. In a hydrogen like atom an electron is orbiting in an orbit having quantum number n . Its frequency of revolution is found to be $13.2 \times 10^{15} \text{ Hz}$. Energy required to free the electron from the atom from the above orbit is 54.4 eV . In time 7 nano-seconds the electron jumps back to orbit having quantum number $\frac{n}{2}$.
If τ be the average torque acted on the electron during the above process, then find $\frac{\tau}{3} \times 10^{27}$ in N-m .
(Given $\frac{h}{\pi} = 2.1 \times 10^{-34} \text{ J-s}$).
Frequency of revolution of electron in the ground state of H-atom $f_0 = 6.6 \times 10^{15} \text{ Hz}$ and ionisation energy of H-atom $E_0 = 13.6 \text{ eV}$)

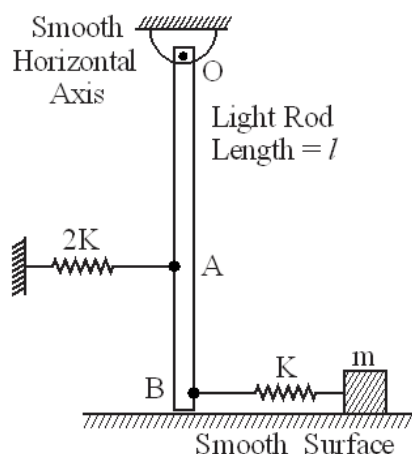
4. As shown in the figure a cylinder of mass M and radius R moves with constant speed v through a region of space that contains dust particles of mass m which are at rest. Number of particles in unit volume is n . The cylinder moves in a direction perpendicular to its axis. Assume that the particles do not interact with each other and that $m \ll M$. All collisions are elastic and surface of the cylinder is smooth. The force per unit length of the cylinder required to maintain the speed v constant is $\frac{K nmRv^2}{3}$. Find the value of K .



5. A positively charged particle is given initial velocity v_0 in XY plane as shown in diagram. Initially the particle was at origin and the region contains uniform, mutually perpendicular electric and magnetic field. Let $\vec{E} = -E\hat{j}$ and $\vec{B} = B\hat{k}$. The minimum magnitude of the magnetic field so that the charge particle crosses the origin again during its subsequent motion is $B_{\min} = \frac{n\pi E}{v_0}$. Find n



6. In the arrangement shown, two springs are attached to a light rod as shown. Length of O to A is $\frac{2\ell}{3}$. System is at equilibrium. The time period of oscillation of the block, if it is displaced slightly is found to be $T = 2\pi\sqrt{\frac{\alpha m}{8K}}$. Find the value of α .



PART-2 : CHEMISTRY
SECTION-I(i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

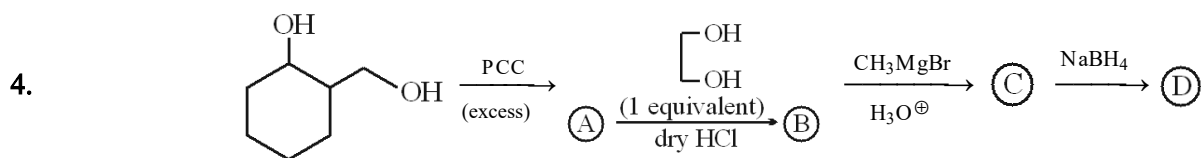
- A metal exists in two allotropic forms, α and β , having a face centred cubic and hexagonal closest packed lattice respectively. Identify the correct statement.
 - Coordination number is higher for the α -form.
 - Density is higher for the β -form.
 - Distance between closest octahedral voids is same for both the forms.
 - Unit cell volume is same for both the forms.
- Consider two Van der waal gases A and B. For gas A, $\lim_{P \rightarrow 0} \left(\frac{\partial Z}{\partial P} \right)_{T=300K} > 0$ and for gas B, $\lim_{P \rightarrow 0} \left(\frac{\partial Z}{\partial P} \right)_{T=300K} < 0$. Identify the incorrect statement.
 - Boyle temperature of gas A < 300 K
 - Boyle temperature of gas B > 300 K.
 - Boyle temperature of both gases is 300 K
 - Critical temperature of B is higher than that of A
- In which of the following cases, the most stable conformer is achiral?
 -
 -
 -
 -

(A) Only IV

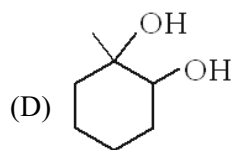
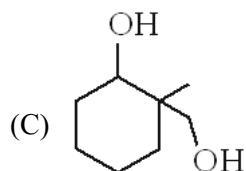
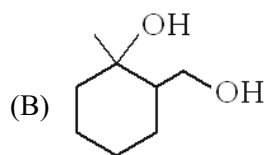
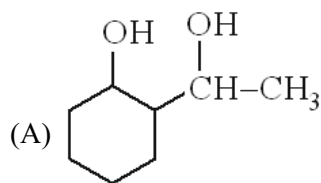
(B) I and IV

(C) I, II and III

(D) In all the cases



product 'D' is:



5. Select the incorrect statement.

- (A) The less stable tautomeric form of H_3PO_3 has three active hydrogens.
- (B) Both H_3PO_3 and H_3PO_2 disproportionate on heating to produce PH_3 and H_3PO_4 .
- (C) In P_4O_{10} all the O-P-O bond angles are identical whereas there are two different P-O bond lengths.
- (D) H_3PO_3 and H_3AsO_3 have different basicity.

6. Which of the following statement is correct regarding extraction of aluminium from bauxite ore:

- (A) In Hall's process CO_2 is formed at the anode.
- (B) Baeyer's process is employed when the bauxite ore is rich in silica.
- (C) In Serpeck's process N_2 is produced due to hydrolysis of aluminium nitride.
- (D) Bauxite ore undergoes roasting before the leaching process.

SECTION-I(ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

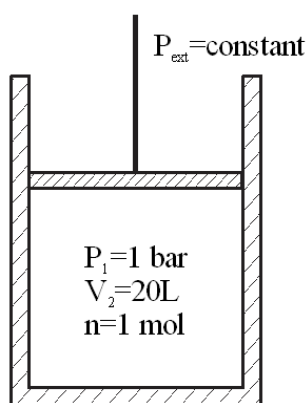
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

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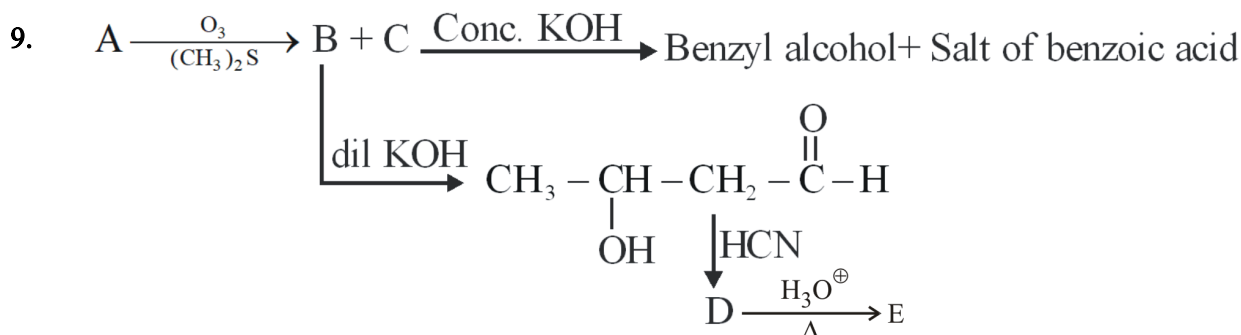
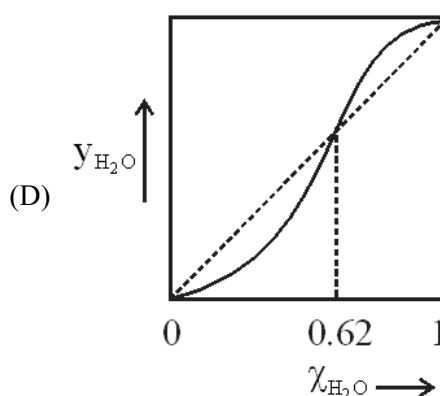
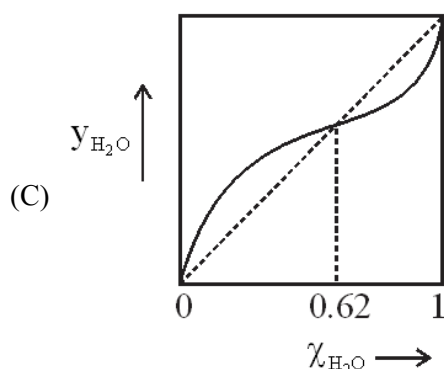
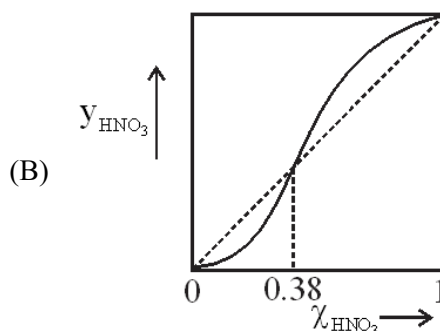
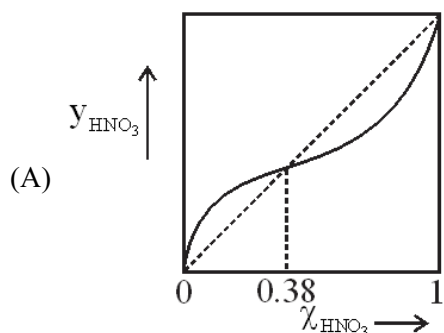
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. 1 mole of He ($C_{v,m} = \frac{3R}{2}$) at 1 bar is compressed adiabatically against a constant external pressure, P_{ext} from an initial volume of 20 L. Assuming He to show ideal gas behaviour, identify the correct statement(s) for the process. (Take $R = 2 \text{ cal mol}^{-1}\text{K}^{-1}$, $\ln 3 = 1.1$, $\ln 5 = 1.6$)

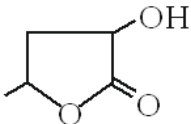


- (A) The lowest possible final volume of the gas after this process is 8L.
- (B) If $P_{\text{ext}} = 3 \text{ atm}$, then $\Delta S = 0.8 \text{ calK}^{-1}$
- (C) ΔS_{sys} is +ve and $\Delta S_{\text{Surroundings}} = 0$ for all values of P_{ext}
- (D) If T_1, V_1 , are the initial temperature and volume of the system, and T_2, V_2 are the final temperature and volume respectively, then $\ln \left(\frac{T_2}{T_1} \right) = \frac{R}{C_v} \ln \left(\frac{V_1}{V_2} \right)$

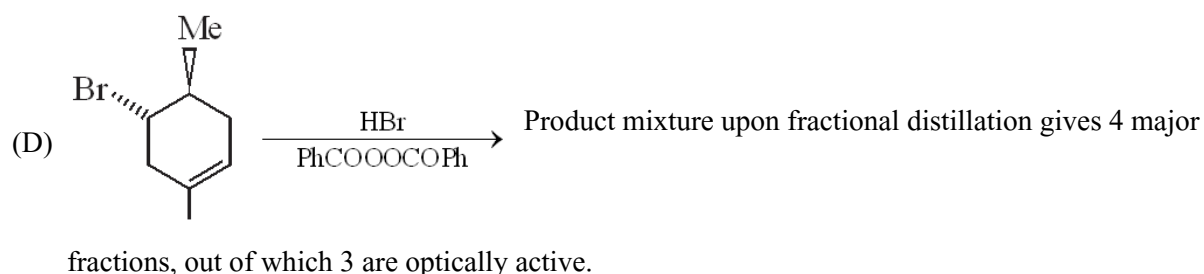
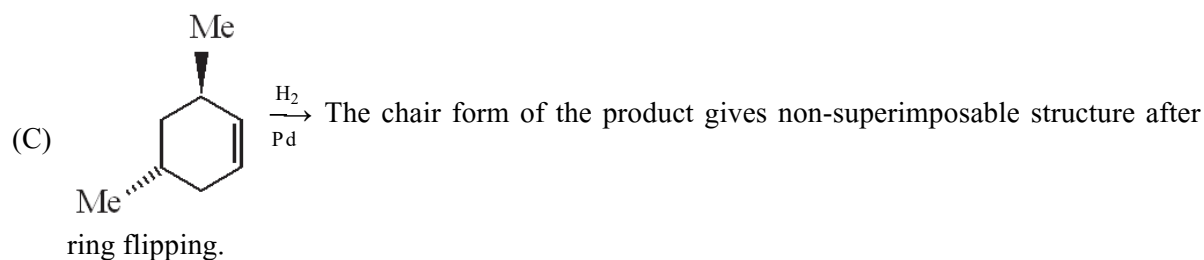
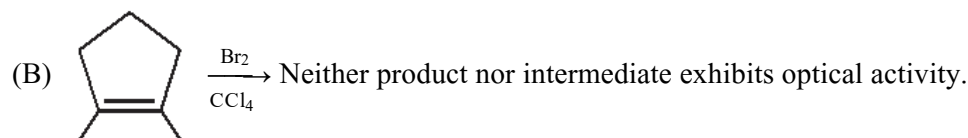
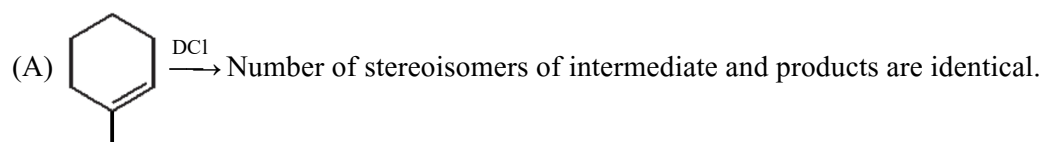
8. An azeotrope of HNO_3 and H_2O is formed with 38 mole percent of HNO_3 . Identify the correct plot(s).
 $y \rightarrow$ vapour phase mole fraction.
 $\chi \rightarrow$ liquid phase mole fraction.



Identify the correct statement(s):

- (A) Structure of 'E' is $\text{CH}_3 - \text{CH} = \text{CH} - \overset{\text{O}}{\parallel} \text{C} - \text{H}$.
- (B) Structure of 'B' and 'C' can be differentiated by Fehling solution.
- (C) Structure of 'E' is 
- (D) Structure of 'A' is $\text{Ph} - \text{CH} = \text{CH} - \text{CH}_3$

10. Which of the following is/are not correctly matched?



11. Select the correct statement(s):

- (A) $[\text{Cr}(\text{NH}_3)_5\text{SO}_4]\text{Cl}$ will produce a white precipitate with AgNO_3 solution but not with BaSO_4 solution.
- (B) Octahedral complexes formed by coordinating strong field ligands with Fe^{+3} will be diamagnetic in nature.
- (C) $[\text{Ni}(\text{en})_3]\text{Cl}_2$, will show optical as well as geometrical isomerism.
- (D) $\text{K}[\text{PtCl}_3\text{C}_2\text{H}_4]$ is an organometallic compound

12. A (is a compound of Ba) and B (is a sulphate) are white crystalline substances, highly soluble in water and decompose at higher temperatures. If an aqueous solution of A (slightly basic) is added to an aqueous solution of B (slightly acidic) a white precipitate C forms. The filtrate is neutral and gives a brown colour solution with an excess of acidified KI solution. When boiled, the filtrate evaporates without the appearance of any residue. White solid D can be prepared by the heating of A in the absence of air. The reaction of D with water gives a colourless solution. This solution, if kept in an open container, slowly precipitates a white solid E and leaves water. Upon prolonged exposure to moist air at room temperature, solid D is transformed into E as well. E can be converted back into D but decomposition above 1400°C is required for this purpose. The reaction of B and D in water forms the precipitate C and is accompanied by a characteristic odour due to formation of a gas. A gives white precipitate with AgNO_3 . Select the correct option(s):
- (A) Compound A is on heating gives neutral, acidic and basic oxide
- (B) Compound B is $(\text{NH}_4)_2\text{SO}_4$.
- (C) Compound E is BaCO_3 .
- (D) I_2 will precipitate as brown coloured.

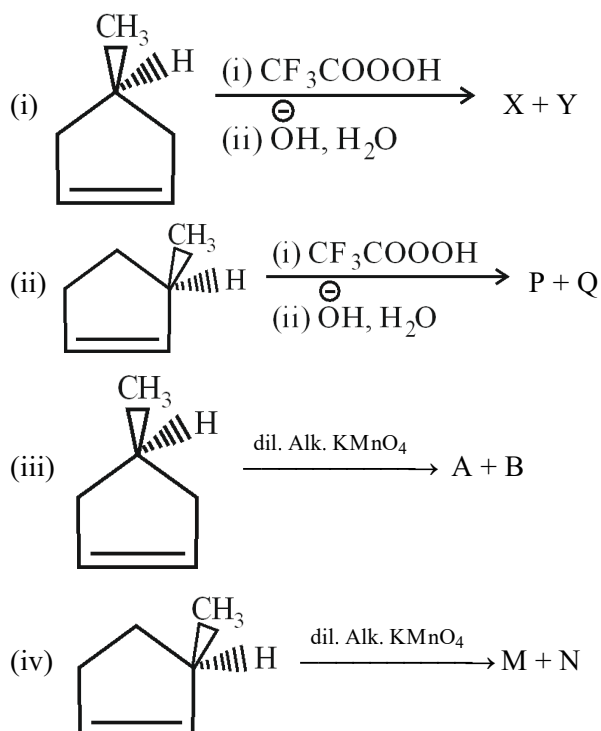
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

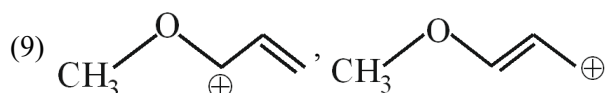
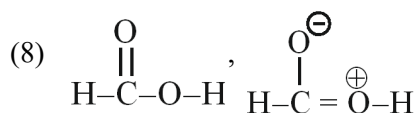
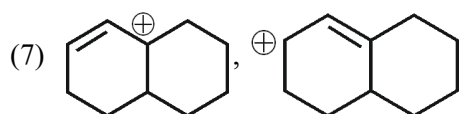
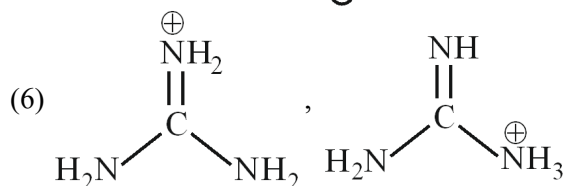
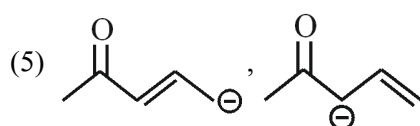
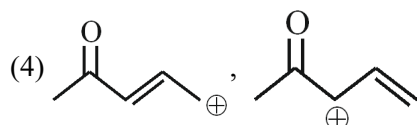
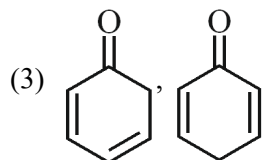
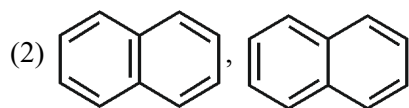
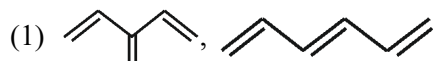
1. Hydrogen peroxide in aqueous solution decomposes by a first order reaction to produce water and oxygen. The rate constant for this decomposition is $7.50 \times 10^{-4} \text{ s}^{-1}$. What quantity of heat (in J) is initially liberated per second from 2.0 L of solution that is 0.02M H_2O_2 ? Given $\Delta_f H^\circ$ of $\text{H}_2\text{O}_2(l) = -187.0 \text{ kJ/mol}$ and $\Delta_f H^\circ \text{H}_2\text{O}(l) = -287.0 \text{ kJ/mol}$.
2. Calculate the pH of solution formed by addition of 1 mole each of HCl, CH_3COONa and HCOONa in enough water to make 1 L solution. The pK_a of HCOOH and CH_3COOH are 4 and 5, respectively.
3. In the given reactions:-



Number of correct statement(s) among the following are :-

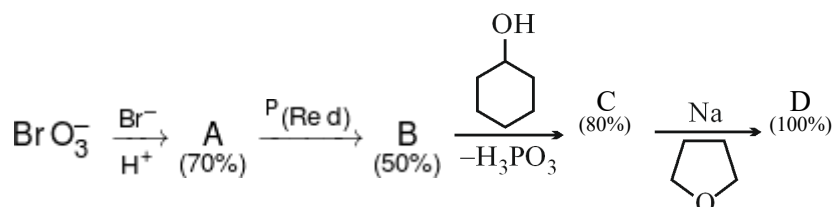
- (1) X and Y are enantiomers
- (2) M and N are enantiomers
- (3) A and B are diastereomers
- (4) (X and Y) are identical with (A and B)
- (5) (P and Q) are identical with (M and N)
- (6) M and N are geometrical isomers

4. In how many of the below options, the first structure is more stable than the second.



5. If the reaction sequence given below is carried out with 4 mol of BrO_3^- and just required amount of Br^- and H^+ , the amount of the product D formed (in g) is _____.

The yields of A, B, C, D are given in the parentheses.



6. The bond angle H-C-H in the formaldehyde molecule is approximately 120° ; the dipole moments for C-H and C=O bonds are $\mu_{\text{C-H}} = 0.4 \text{ D}$ and $\mu_{\text{C=O}} = 2.3 \text{ D}$, respectively. The net dipole moment of the formaldehyde is $X \times 10^{-Y} \text{ Cm}$ (in scientific notation). The value of X is _____.

PART-3 : MATHEMATICS
SECTION-I(i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

-
1. If α and β are exactly two distinct real roots of the equation $(x^2 + 5x - b^2)(2x^2 - ax - 6) = 0$ where $a, b \in \mathbb{R}$, (Given that equation has only real roots) then
- (A) $\alpha^2 + \beta^2 = 30$
- (B) $a^2 + b^2 = 105$
- (C) $\alpha^2 + \beta^2 = 19$
- (D) $a^2 + b^2 + \alpha^2 + \beta^2 = 134$
2. If $f(x) = x^4 - 3x^2 + 7x - 1$ and $g(x) = \cot^{-1}(2x^2) + \sin^{-1}\left(\frac{1}{1+x^2}\right)$ are two functions, then number of integers in the range of $\text{gof}(x)$ is
- (A) 1
- (B) 2
- (C) 3
- (D) 4
3. $\lim_{x \rightarrow 0^+} \frac{(1 + \tan x)^{2 \cot x} - e^{2 \cos x}}{1 - \cos(\ell n(1 + \sin \sqrt{x}))}$
- (A) $2e^2$
- (B) $2e$
- (C) $-2e$
- (D) $-2e^2$

4. $\int e^x \left[\frac{2 \sin x}{(\sin x + \cos x)^2} - \frac{2 \cos x}{(\sin x - \cos x)^2} \right] dx$

where 'C' is the constant of integration.

(A) $e^x \left[\frac{2 \sin x}{\sin^2 x - \cos^2 x} \right] + C$

(B) $e^x \left[\frac{2 \cos x}{\sin^2 x - \cos^2 x} \right] + C$

(C) $e^x \left(\frac{\sin^2 x - \cos^2 x}{2 \sin x} \right) + C$

(D) $e^x \left(\frac{\sin^2 x - \cos^2 x}{2 \cos x} \right) + C$

5. The value of $S = (\sin^3 \theta + \cos^3 \theta) + (\sin^3 3\theta + \cos^3 3\theta) + (\sin^3 5\theta + \cos^3 5\theta) + \dots + (\sin^3 19\theta + \cos^3 19\theta)$ when $\theta = \frac{\pi}{12}$ is $\frac{-a}{\sqrt{b}}$ where a & b are relatively prime numbers, then $|b - 6a|$ equals

(A) 1

(B) 2

(C) 3

(D) 4

6. Consider $f(x) = x^3 + ax^2 + bx + c$ where $a, b, c \in \{1, 2, 3, 4, 5, 6\}$. The probability that $f(x) = 0$ has all real roots given that $f(x)$ is non-decreasing function, is

(A) $\frac{1}{216}$

(B) $\frac{1}{96}$

(C) $\frac{1}{108}$

(D) $\frac{1}{36}$

SECTION-I(ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. The vertices of triangle ABC are A(2, 0, 2); B(-1, 1, 1) and C(1, -2, 4). The point D and E divide AB and CA in the ratio 1 : 2 respectively. Another point F is taken in space such that perpendicular drawn from F to the plane containing $\triangle ABC$, meets the plane at the point of intersection (point P) of the line segment CD & BE. If the distance of F from the plane of triangle ABC is $\sqrt{2}$ units then

(A) $\frac{\text{area of } \triangle ABC}{\text{area of } \triangle PBC} = \frac{7\sqrt{3}}{6}$

(B) Volume of tetrahedral ABCF = $\frac{7}{3}$ cubic units

(C) Equation of line AF is $\vec{r} = 2(\hat{i} + \hat{k}) + \lambda(\hat{i} + 2\hat{j})$

(D) Equation of line AF is $\vec{r} = 2(\hat{i} + \hat{k}) + \lambda(\hat{i} - 2\hat{k})$

8. If chords of the ellipse $\frac{x^2}{a^2} + y^2 = 2$, passing through (a, -1) are bisected by the line $x + y = 1$, then

(A) least positive integral value of a is 2.

(B) least positive integral value of a is 1.

(C) sum of integral values of a in $(-10, 5)$ is -8.

(D) sum of negative integral values of a in $(-20, -9)$ is -145.

9. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function satisfying $f(x+y) = f(x) + f(y) + (x+y)e^{x+y} - xe^x - ye^y + 4xy$ $\forall x, y \in \mathbb{R}$ and $f'(0) = 1$, then which of the following option(s) is/are **correct**?
- (A) $f(x) = e^{-x+3}$; $x > 0$ has exactly one solution
- (B) $f(2) - f'(2) < 0$
- (C) $3f(x) = \operatorname{sgn}(-x^2 + 3x - 6)$ has exactly one solution in $x \in [2, 5]$
- (D) $f(3) - 3e^3 = x^2 - 6x + 5$ has exactly two solution
10. $\int_0^{\pi} [\{x\} + [\cot x]] dx$ is greater than or equal to
- [Note: $\{z\}$ & $[z]$ denotes fractional part function and greatest integer function of z .]
- (A) 0
- (B) 1
- (C) $-\frac{\pi}{2}$
- (D) $-\pi$
11. Let OABC is a rhombus where O is origin such that vertices A (z_1) and C (z_3) lie on the circle $|z| = 2$ and the vertex B (z_2) lie on the circle $|z| = 3$. Let $\operatorname{Arg}(z_2 - z_1) = \frac{\pi}{3}$, then
- (A) $z_1 = \left(\frac{3 - \sqrt{7}i}{3 + \sqrt{7}i} \right) (1 + \sqrt{3}i)$
- (B) $z_2 = \left(\frac{6 + 6\sqrt{3}i}{3 - \sqrt{7}i} \right)$
- (C) $z_3 = (1 + \sqrt{3}i)$
- (D) $z_1 = \left(\frac{3 - \sqrt{7}i}{3 + \sqrt{7}i} \right) (1 - \sqrt{3}i)$
12. Let $f(x) = -x^3 + px^2 + qx + 6\operatorname{sgn}(x^2 + 2x + 5)$ where $p, q \in \mathbb{R}$ & sgn denotes signum function. If the largest possible interval in which $f'(x)$ is positive is $(-2, 1)$, then $(p + q)$ is less than or equal to
- (A) 4
- (B) 5
- (C) 6
- (D) 9

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Let a plane P containing a line $\vec{r} = (\hat{i} + \hat{j} + \hat{k}) + \lambda (-\hat{i} + 3\hat{j} + 2\hat{k})$ and perpendicular to $2x + 3y - z = 2$. If the area of projection triangle formed on plane P whose vertices are $A_r = 2r\hat{i} + 3(-1)^r\hat{j} + r^2\hat{k}$ where $r \in \mathbb{N}$ & $1 \leq r \leq 3$ is $a\sqrt{b}$ where a, b are relatively prime numbers, then $\left(\frac{b}{a}\right)$ is equal to
2. In a book with page numbers 1 to n, four consecutive pages are torn off and the sum of the numbers on the remaining pages is 1596, then find the sum of possible values of n. (4 consecutive pages means page numbers m, m+1, m+2 & m+3, $m \in \mathbb{N}$)
3. From a fixed point A, three normals are drawn to the parabola $y^2 = 4ax$ at the points P, Q and R. Two circles C_1 and C_2 be drawn on AP and AQ as diameter. If slope of the common chord of the circles C_1 and C_2 be m_1 and the slope of the tangent to the parabola at the point R be m_2 , then $m_1 \times m_2$ is equal to
4. Let S be the set containing all matrices $A = [a_{ij}]_{3 \times 3}$ having entries from the set $\{0, 1, 2, 4\}$ such that sum of all principal diagonal entries of $AA^T = 17$ then number of matrices in S is
5. If $\int_0^1 ((6x^4 - f(x)) f(x) dx = 1$, then the area of region enclosed by $y = f(x)$, $x = 0$ and $x = 2$.
6. Let $u = \sin^7 x$, $v = \cos x$. If $\frac{d^2}{dx^2} \left(\frac{d^3 u}{dv^3} \right) \bigg|_{x=\frac{\pi}{8}} = 2^{\lambda_1} \cdot 3^{\lambda_2} \cdot 5^{\lambda_3} \cdot 7^{\lambda_4}$, then the value of $(\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4)$, is

Space for Rough Work

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